Tydenso

Symbolic tensor algebra inside Typst

User manual · version 0.1.0

Build indexed tensors as readable Typst values, transform them with Idenso, and print them with Spenso's own notation.

Repository · MIT license · Spenso Python API · Idenso Python API

Contents

1 Start with a contraction

Tydenso is the tensor-focused companion to Tymbolica. It has its own package, manual, printer, and WebAssembly engine. You do not need Tymbolica to construct, simplify, inspect, or display a tensor expression.

The first example contracts a Minkowski metric with a vector. A representation is an ordinary Typst dictionary; `slot` turns an index label into another dictionary; `vector` creates a callable rank-one tensor name.

```
#let V = mink(4)
#let mu = slot(V, "mu")
#let nu = slot(V, "nu")
#let p = vector("p")

#let before = math($#metric(V, mu, nu) #p(nu)$)
#let after = simplify-metrics(before)

$
  #to-typst(before)
  quad arrow.r.long
  #to-typst(after)
$
```

$$p^{\nu}\, g^{\mu\nu} \quad \longrightarrow p^{\mu}$$

The constructor calls already produce readable math. Their hidden, versioned metadata contains the exact Symbolica Atom, so `math` does not have to infer a tensor from its appearance. It returns Atom bytes for algebra; `to-typst` renders transformed results.

1.1 Installation

Until Tydenso is published separately, install the `tydenso` directory from this repository as its own local package. On Linux, place or symlink it at:

`~/.local/share/typst/packages/local/tydenso/0.1.0`

Then import it independently:

```
#import "@local/tydenso:0.1.0": *
```

From a checkout, `#import "path/to/tymbolica/tydenso/lib.typ": *` works as well. The package directory already contains its compressed engine and loader.

1.2 How this corresponds to the Python API

The public concepts follow Spenso's Python surface, adapted to what is natural in Typst. Python objects can overload calls; Typst dictionaries cannot, so index construction is the explicit `slot(V, "mu")`. Tensor names are functions, which keeps the pleasant `p(mu)` spelling.

Python	Typst

`V = Representation.mink(4)`
`mu = V("mu")`
`nu = V("nu")`
`p = TensorName("p")`
`expr = TensorName.g(mu, nu) * p(nu)`
	`#let V = mink(4)`
	`#let mu = slot(V, "mu")`
	`#let nu = slot(V, "nu")`
	`#let p = vector("p")`
	`#let expr = math($#metric(V, mu, nu)`
	`#p(nu)$)`

Representations and slots stay transparent dictionaries. Symbols and tensor calls are visible Typst math with exact Atom metadata; completed algebraic expressions are Atom bytes.

2 Build tensor expressions

2.1 Representations and slots are data

Built-in constructors cover the representations initialized by Spenso and Idenso: `mink`, `euc`, `lor`, `bis`, `spf`, `cof`, `coad`, and `cos`. Use `representation` when a model introduces another representation.

A custom representation can name its automatic indices. The palette repeats; each pass adds a subscript. Manually written indices use the same notation but keep their own symbolic identity.

```
#let M = representation(
  "M",
  4,
  namespace: "example",
  self-dual: true,
  indices: ($mu$, $nu$),
)
#let T = tensor("T", namespace: "example")
#let expression = T(
  slot(M, 1),       // mu
  slot(M, 2),       // nu
  slot(M, 3),       // mu_1
  slot(M, $rho_2$), // exactly the index written here
)

#let detailed = print-settings(with-dim: true)
$ #to-typst(expression, settings: detailed) $
```

$$T^{\mu^{M_4}\nu^{M_4}\mu_1^{M_4}\rho_2^{M_4}}$$

Here the qualified form distinguishes the indices as members of M_4. Leave `indices` unset to keep the ordinary numeric display used by the built-in representations.

```
#let color = cof(3)
#let a = slot(color, "a")
#let b-lower = slot(color, "b", dual: true)

#table(
  columns: (auto, 1fr),
  inset: 5pt,
  [representation], [#color.name],
  [dimension], [#color.dimension],
  [first index], [#a.index],
  [second index is dual], [#b-lower.dual],
)
```

representation	cof
dimension	3
first index	a
second index is dual	true

Because those values are dictionaries, document code can validate dimensions, generate index families in a loop, or attach its own metadata before asking the plugin to construct an Atom. `dual-representation` returns the paired representation; `metric`, `identity-tensor`, and `flat-tensor` accept either index labels or already constructed slots.

2.2 Tensor names carry symmetry

`tensor` mirrors the useful part of Spenso's `TensorName`: a name plus optional symmetric, antisymmetric, cyclic, or linear behavior. Symbolica applies those attributes while it constructs the function, not as a later cosmetic step.

```
#let V = euc(3)
#let mu-slot = slot(V, "mu")
#let nu-slot = slot(V, "nu")
#let F = tensor("F", antisymmetric: true)

#let cancellation = math($#F(mu-slot, nu-slot) + #F(nu-slot, mu-slot)$)
$ F^(mu nu) + F^(nu mu) = #to-typst(cancellation) $
```

$$F^{\mu\nu} + F^{\nu\mu} = 0$$

Only one of `symmetric`, `antisymmetric`, and `cycle-symmetric` may be true for one tensor name. The `linear` flag is independent.

2.3 Write tensor algebra as math

Tensor and vector constructors are Typst functions. Interpolate their calls in math with `#F(...)`; interpolate scalar symbols with `#mass`. Parsely reads the ordinary arithmetic, while each annotated value contributes its exact Atom.

```
#let V = mink("D")
#let mu = slot(V, "mu")
#let nu = slot(V, "nu")
#let p = vector("p")
#let mass = symbol("m", namespace: "model")

#let shell = math($#metric(V, mu, nu) #p(mu) #p(nu) - #mass^2$)
$ #to-typst(shell) $
```

$$p^{\mu}p^{\nu}g^{\mu\nu} - m^{2}$$

The structural `add`, `mul`, `neg`, `sub`, `div`, and `pow` functions remain useful for generated expressions. They accept Atom bytes, annotated content, numbers, slots, and representation dictionaries. Both paths construct the same Atom; neither reconstructs tensors from printed subscripts.

2.4 Build open and closed Spenso chains

The chain helpers construct Spenso's actual Atom heads; they are not drawing commands. Compact endpoints are ordinary rank-one tensors carrying a representation, while `gamma(mu)` supplies Spenso's `in` and `out` placeholders for a chain factor.

```
#let M = mink(4)
#let B = bis(4)
#let mu = slot(M, "mu")
#let nu = slot(M, "nu")

#let p = vector("p")
#let q = vector("q")
#let u = vector("u")
#let v = vector("v")

#let scalar = dot(p(1, M), q(2, M))
#let open = chain(
  u(1, B),
  v(2, B),
  gamma(mu),
  gamma(p(1, M)),
  gamma(nu),
)
#let closed = trace(
  B,
  cyclic(gamma(mu), gamma(p(1, M)), gamma(nu)),
)

$ #to-typst(scalar) $
$ #to-typst(open) $
$ #to-typst(closed) $
```

$$p_{1} \cdot q_{2}$$

$$\langle\, u_{1} \mid \left[\gamma^{\mu} \not{p}_{1} \gamma^{\nu}\right] \mid v_{2} \,\rangle$$

$$\mathrm{Tr}\left(\gamma^{\mu} \not{p}_{1} \gamma^{\nu}\right)$$

Use `gamma(mu, a, b)` when the bispinor slots are explicit. `chain` keeps an ordered open sequence; `cyclic` marks the factor list of a closed sequence; and `trace` combines that cycle with its representation. The standalone notation example also inspects the constructed trees and checks their exact Spenso shapes.

3 Transform tensors

Idenso provides the domain-specific transformations. The most common ones simplify metrics, gamma matrices, and color structures; selective expanders and index-wrapping functions are available for lower-level workflows.

3.1 Contract a metric chain

The next expression contains two metrics and one vector. Simplifying metrics eliminates both contracted dummy indices in one pass.

```
#let V = mink(4)
#let mu = slot(V, "mu")
#let nu = slot(V, "nu")
#let rho = slot(V, "rho")
#let p = vector("p")

#let chain = mul(
  metric(V, mu, nu),
  metric(V, nu, rho),
  p(rho),
)
#let reduced = simplify-metrics(chain)

$
  #to-typst(chain)
  quad arrow.r.long
  #to-typst(reduced)
$
```

$$p^{\rho} g^{\mu\nu} g^{\nu\rho} \quad \longrightarrow p^{\mu}$$

`list-dangling` returns the free indices as Atom payloads. Pass an element to `to-typst`, `to-string`, or `inspect` just like any other expression.

3.2 Know which family to call

Family	Purpose
`simplify-metrics`, `expand-metrics`, `expand-mink`, `expand-bis`	Metric, Minkowski, and bispinor structure.
`simplify-gamma`, `dirac-adjoint`	Dirac chains and conjugation.
`simplify-color`, `expand-color`	Color generators and structure constants.
`cook-function`, `cook-indices`, `wrap-dummies`, `wrap-indices`	Canonical index organization and explicit wrappers.
`to-dots`	Rewrite supported contractions as dot products.

The mathematical conventions and supported identities track Idenso's API.

4 Print and inspect

4.1 Use the Spenso printer directly

Tydenso does not send tensor output through Tymbolica's general printer. `to-typst-source` uses Symbolica's Typst arithmetic mode together with Spenso's custom tensor notation. `to-string` uses Spenso's compact Symbolica notation.

```
#let V = mink(4)
#let p = vector("p")
#let expression = p(1, slot(V, "mu"))

#let detailed = print-settings(with-dim: true, commas: true)

Rendered: $ #to-typst(expression, settings: detailed) $

Compact Spenso form:
#raw(to-string(expression), block: true)
```

```
Rendered:
```

$$p_{1}{}^{\mu^{mink_4}}$$

```
Compact Spenso form:
p(1,αmu)
```

Upper and lower indices align in matching columns. A plain self-dual slot is placed on the top row; for a dualizable representation, its dual orientation is placed on the bottom row.

`print-settings` exposes the real `SpensoPrintSettings` switches: `with-dim`, `parens`, `commas`, `index-subscripts`, and `symbol-scripts`. Start from either the `"typst"` or `"compact"` preset and override only what the document needs.

4.2 Inspect the Atom as CBOR data

Atom bytes remain the lossless transformation and interchange format. Before a transformation, annotated constructor content carries those same bytes in metadata. `inspect` accepts either form and decodes it to recursive Typst data. Every node has a `kind`; function nodes also expose their full and short names, arguments, and symmetry flags.

```
#let V = euc(3)
#let A = tensor("A", symmetric: true)
#let expression = A(slot(V, "i"), slot(V, "j"))
#let tree = inspect(expression)

#table(
  columns: (auto, 1fr),
  inset: 5pt,
  [node kind], [#tree.kind],
  [function], [#tree.short-name],
  [arguments], [#tree.arguments.len()],
  [symmetric], [#tree.symmetric],
)
```

node kind	function
function	A
arguments	2
symmetric	true

> **Why keep both forms?** CBOR is convenient for layouts, diagnostics, and package-level APIs. Reconstructing algebra from that tree would lose Symbolica state and exact coefficient details, so transformations continue to consume Atom bytes.

5 Work alongside Tymbolica

Tydenso and Tymbolica share the same Atom payload. A package can construct and simplify tensors with Tydenso, then pass the result to Tymbolica for a general algebraic operation. Custom representations keep the information Tydenso needs to use them again, including their index palette.

```
#import "@local/tydenso:0.1.0" as tensors
#import "@local/tymbolica:0.1.0" as algebra

#let V = tensors.representation(
  "M",
  4,
  namespace: "model",
  self-dual: true,
  indices: ($mu$, $nu$),
)
#let p = tensors.vector("p")
```

```
#let expression = p(tensors.slot(V, 1))

#let expanded = algebra.expand(expression)
#tensors.to-typst(expanded)
```

Keep package versions aligned. Matrix payloads from Tymbolica are a different format and are not Tydenso inputs.

6 API reference

The groups below cover the imported top-level API. `init` returns the same surface as a dictionary bound to a selected plugin module.

6.1 Tensor construction

6.1.1 tensor

Construct a named tensor function.

The result is callable with any mixture of slots, representations, scalar values, and compatible Atom payloads. Symmetry flags follow Spenso's `TensorName` model and are registered on the underlying Symbolica symbol.

```
#let V = mink(4)
#let Z = tensor("Z")
#to-typst(Z(slot(V, "mu"), slot(V, "nu")))
```

$$Z^{\mu\nu}$$

Parameters

```
tensor(
  name: str,
  namespace: str,
  symmetric: bool,
  antisymmetric: bool,
  cycle-symmetric: bool,
  linear: bool,
  tags: array
) -> function
```

6.1.1.0.1 name `str`

Tensor name.

6.1.1.0.2 namespace `str`

Symbol namespace.

Default: `"spenso"`

6.1.1.0.3 symmetric `bool`

Make the tensor symmetric under argument permutation.

Default: `false`

6.1.1.0.4 antisymmetric `bool`

Make the tensor antisymmetric under argument permutation.

Default: `false`

6.1.1.0.5 cycle-symmetric `bool`

Make the tensor symmetric under cyclic argument permutation.

Default: `false`

6.1.1.0.6 linear `bool`

Mark the tensor function as linear.

Default: `false`

6.1.1.0.7 tags `array`

Portable semantic labels retained in attached metadata.

Default: `()`

6.1.2 vector

Construct a named rank-one tensor function.

Non-slot arguments are rendered as symbol subscripts in Spenso's Typst mode, while the vector slot is rendered as an abstract index.

```
#let V = mink(4)
#let p = vector("p")
#to-typst(p(1, slot(V, "mu")))
```

$p_1{}^{\mu}$

Parameters

```
vector(
  name: str,
  namespace: str,
  linear: bool,
  tags: array
) -> function
```

6.1.2.0.1 name `str`

Vector name.

6.1.2.0.2 namespace `str`

Symbol namespace.

Default: `"spenso"`

6.1.2.0.3 linear `bool`

Mark the vector function as linear.

Default: `false`

6.1.2.0.4 tags `array`

Portable semantic labels retained in attached metadata.

Default: `()`

6.1.3 symbol

Construct a symbolic scalar with exact Atom metadata.

Parameters

```
symbol(
  name: str,
  namespace: str,
  tags: array
) -> content
```

6.1.3.0.1 name `str`

Symbol name.

6.1.3.0.2 namespace `str`

Symbol namespace.

Default: `"spenso"`

6.1.3.0.3 tags `array`

Portable semantic labels retained in attached metadata.

Default: `()`

6.1.4 function

Construct a callable symbolic function with exact Atom metadata.

This is distinct from `symbol`: a Typst content value is not callable. Calling the returned function annotates the complete function call.

Parameters

```
function(
  name: str,
  namespace: str,
  tags: array
) -> function
```

6.1.4.0.1 name `str`

Function-head name.

6.1.4.0.2 namespace `str`

Symbol namespace.

Default: `"spenso"`

6.1.4.0.3 tags `array`

Portable semantic labels retained in attached metadata.

Default: `()`

6.1.5 representation

Describe a representation and attach its tensor-building methods.

The returned dictionary is inspectable Typst data. Its `slot`, `metric`, `identity`, `flat`, and `dual` fields are functions.

Parameters

```
representation(
  name: str,
  dimension: int str bytes,
  namespace: str,
  self-dual: bool,
  is-dual: bool,
  dual-name: str none,
  indices: array none,
  index-start: int,
  index-row: str none
) -> dictionary
```

6.1.5.0.1 name `str`

Symbolic representation name.

6.1.5.0.2 dimension `int` or `str` or `bytes`

Dimension, which may itself be symbolic.

6.1.5.0.3 namespace `str`

Symbol namespace.

Default: `"spenso"`

6.1.5.0.4 self-dual `bool`

Whether the representation is its own dual.

Default: `false`

6.1.5.0.5 is-dual `bool`

Whether this value denotes the dual orientation of a non-self-dual representation.

Default: `false`

6.1.5.0.6 dual-name `str` or `none`

Deprecated compatibility parameter. A representation and its dual must use the same canonical symbol; omit this or set it equal to `name`.

Default: `none`

6.1.5.0.7 indices `array` or `none`

Fixed cyclic display palette for automatic integer indices. Entries may be Typst math content, strings, or integers. `none` keeps numeric display.

Default: `none`

6.1.5.0.8 index-start `int`

Integer represented by the first palette entry. Each wrap adds a numeric subscript, so `($mu$, $nu$)` maps 1, 2, 3 to μ, ν, μ_1.

Default: `1`

6.1.5.0.9 index-row `str` or `none`

Preferred Typst script row for the representation's base orientation. `none` selects bottom for the built-in `spenso::bis` representation and top for every other representation. A dualizable representation's dual orientation uses the opposite row.

Default: `none`

6.1.6 mink

Construct a Minkowski representation.

Parameters

`mink(dimension) -> dictionary`

6.1.7 euc

Construct a Euclidean representation.

Parameters

`euc(dimension) -> dictionary`

6.1.8 lor

Construct a Lorentz representation.

Parameters

`lor(dimension) -> dictionary`

6.1.9 bis

Construct a bispinor representation.

Parameters

```
bis(dimension) -> dictionary
```

6.1.10 spf

Construct a spin-fundamental representation.

Parameters

```
spf(dimension) -> dictionary
```

6.1.11 cof

Construct a color-fundamental representation.

Parameters

```
cof(dimension) -> dictionary
```

6.1.12 coad

Construct a color-adjoint representation.

Parameters

```
coad(dimension) -> dictionary
```

6.1.13 cos

Construct a color-sextet representation.

Parameters

```
cos(dimension) -> dictionary
```

6.1.14 slot

Construct an indexed slot from a representation.

The result is an ordinary dictionary, so its representation, index, and variance remain visible to Typst before an expression is constructed.

Parameters

```
slot(
  representation: dictionary,
  index: int str bytes content,
  dual: bool none
) -> dictionary
```

6.1.14.0.1 representation `dictionary`

Representation dictionary.

6.1.14.0.2 index `int` or `str` or `bytes` or `content`

Abstract index label. Typst math content such as `$mu_1$` is retained as safe display metadata on a symbolic index.

6.1.14.0.3 dual `bool` or `none`

Wrap the slot with Spenso's dual-index marker. `none` inherits the representation's `is-dual` field.

Default: `none`

6.1.15 metric

Construct the metric tensor for a representation.

Index labels are converted to slots automatically; existing slots pass through unchanged.

Parameters

```
metric(
  representation,
  first,
  second
) -> content
```

6.1.16 identity-tensor

Construct the identity tensor for a representation.

Parameters

```
identity-tensor(
  representation,
  first,
  second
) -> content
```

6.1.17 flat-tensor

Construct Spenso's musical isomorphism tensor for a representation.

Parameters

```
flat-tensor(
  representation,
  first,
  second
) -> content
```

6.1.18 dual-representation

Return the representation paired with this representation.

Parameters

```
dual-representation(representation) -> dictionary
```

6.2 Expression construction

6.2.1 math

Parse Typst math into an exact Symbolica Atom payload.

Annotated values created by `symbol`, `function`, `tensor`, `vector`, and the named tensor constructors are imported from their exact Atom metadata. Use interpolation for callable bindings, for example `#F(mu, nu)`.

Parameters

```
math(
  equation: content,
  grammar: dictionary none,
  namespace: str none
) -> bytes
```

6.2.1.0.1 equation `content`

Math content, normally written as `$...$`.

6.2.1.0.2 grammar `dictionary` or `none`

Parser grammar override. `none` uses the engine grammar.

Default: `none`

6.2.1.0.3 namespace `str` or `none`

Namespace for unannotated parsed symbols.

Default: `none`

6.2.2 atom

Convert a supported value or annotated math expression to Atom bytes.

Parameters

```
atom(value) -> bytes
```

6.2.3 add

Add symbolic expressions or constructor values.

Parameters

```
add(..terms) -> bytes
```

6.2.4 mul

Multiply symbolic expressions or constructor values.

Parameters

```
mul(..factors) -> bytes
```

6.2.5 neg

Negate a symbolic expression or constructor value.

Parameters

```
neg(expression) -> bytes
```

6.2.6 sub

Subtract two symbolic expressions or constructor values.

Parameters

```
sub(
  left,
  right
) -> bytes
```

6.2.7 div

Divide two symbolic expressions or constructor values.

Parameters

```
div(
  numerator,
  denominator
) -> bytes
```

6.2.8 pow

Raise a symbolic expression or constructor value to a power.

Parameters

```
pow(
  base,
  exponent
) -> bytes
```

6.3 Spenso products and chains

6.3.1 dot

Construct Spenso's compact scalar product `dot(left, right)`.

Rank-one tensor calls with a representation and no explicit slot are the compact Schoonschip form. Thus `dot(p(M), q(M))` emits exactly `dot(p(mink(4)),q(mink(4)))` when `M = mink(4)`.

Parameters

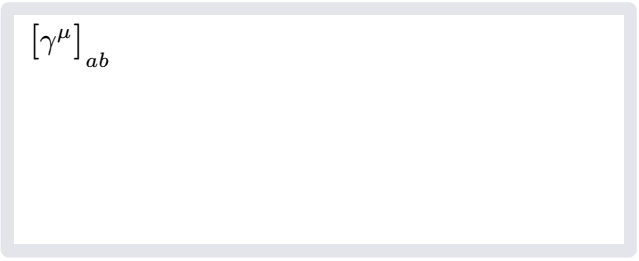
```
dot(
  left,
  right
) -> content
```

6.3.2 gamma

Construct an Idenso gamma tensor or a gamma chain factor.

With one argument this emits the actual Spenso factor `gamma(in,out,lorentz)`. With all three arguments it emits `gamma(first,second,lorentz)`, which is the explicit tensor order stored in the Atom even though the Typst API places the Lorentz argument first.

```
#let M = mink(4)
#let B = bis(4)
#let mu = slot(M, "mu")
#let a = slot(B, "a")
#let b = slot(B, "b")
#let factor = gamma(mu)
#let explicit = gamma(mu, a, b)
#to-typst(chain(a, b, factor))
```



Parameters

```
gamma(
  lorentz,
  ..endpoints
) -> content
```

6.3.3 chain

Construct an open Spenso chain.

`start` and `end` may be explicit slots or compact rank-one tensors. For example, if `u = vector("u")` and `B = bis(4)`, then `u(B)` is the actual compact endpoint Atom `u(bis(4))`; no `bra` or `ket` head is introduced. Factors such as `gamma(mu)` already contain Spenso's `in` and `out` placeholders.

Parameters

```
chain(
  start,
  end,
  ..factors
) -> content
```

6.3.4 cyclic

Construct Spenso's inert cycle-symmetric projector.

This emits the actual Atom `cyclic(factors...)`.

Parameters

```
cyclic(..factors) -> content
```

6.3.5 trace

Construct a canonical closed Spenso chain.

A non-empty call accepts one explicit `cyclic(...)` payload and therefore emits exactly `trace(rep,cyclic(factors...))`. Passing a raw factor list is rejected instead of being silently rewritten. An empty call emits `trace(rep)`.

```
#let M = mink(4)
#let B = bis(4)
#let mu = slot(M, "mu")
#let nu = slot(M, "nu")
#to-typst(trace(B, cyclic(gamma(mu),
gamma(nu))))
```

$$\mathrm{Tr}\left(\gamma^{\mu}\gamma^{\nu}\right)$$

Parameters

```
trace(
  representation,
  ..factors
) -> content
```

6.4 Printing and inspection

6.4.1 print-settings

Create Spenso printer settings.

`"typst"` uses valid Typst math syntax with indexed tensor notation; `"compact"` uses Spenso's concise Symbolica-style form. Any field may be overridden without changing the others.

Parameters

```
print-settings(
  preset: str,
  with-dim: bool none,
  parens: bool none,
  commas: bool none,
  index-subscripts: bool none,
  symbol-scripts: bool none
) -> dictionary
```

6.4.1.0.1 preset `str`

Base preset: `"typst"` or `"compact"`.

Default: `"typst"`

6.4.1.0.2 with-dim `bool` or `none`

Include representation dimensions in printed slots.

Default: none

6.4.1.0.3 parens `bool` or `none`

Use parentheses around tensor arguments.

Default: none

6.4.1.0.4 commas `bool` or `none`

Separate tensor arguments with commas.

Default: none

6.4.1.0.5 index-subscripts `bool` or `none`

Render indices as subscripts where supported.

Default: none

6.4.1.0.6 symbol-scripts `bool` or `none`

Render non-slot tensor arguments as symbol scripts where supported.

Default: none

6.4.2 to-typst-source

Print an expression as Typst math source using Spenso's printer.

Parameters

```
to-typst-source(
  expression,
  settings: dictionary
) -> str
```

6.4.2.0.1 expression

Atom payload or constructor value.

6.4.2.0.2 settings `dictionary`

Settings created by `print-settings`.

Default: `_print-settings()`

6.4.3 to-typst

Print and evaluate an expression as Typst math content.

Parameters

```
to-typst(
  expression,
  settings: dictionary,
  block: bool
) -> content
```

6.4.3.0.1 expression

Atom payload or constructor value.

6.4.3.0.2 settings `dictionary`

Settings created by `print-settings`.

Default: `_print-settings()`

6.4.3.0.3 block `bool`

Display the result as a block equation.

Default: `false`

6.4.4 to-string

Print an expression in Spenso's compact Symbolica notation.

Parameters

```
to-string(
  expression,
  settings: dictionary
) -> str
```

6.4.4.0.1 expression

Atom payload or constructor value.

6.4.4.0.2 settings `dictionary`

Settings created by `print-settings`.

Default: `_print-settings(preset: "compact")`

6.4.5 inspect

Decode an Atom payload into a recursive CBOR expression tree.

The result uses `kind` values such as `symbol`, `number`, `function`, `sum`, `product`, and `power`. Function nodes include their name, arguments, and symmetry flags. This view is ordinary Typst data; transforming the expression still uses its lossless Atom payload.

Parameters

```
inspect(expression) -> dictionary
```

6.5 Metric and index transformations

6.5.1 cook-function

Cook a tensor function into Idenso's canonical structure.

Parameters

```
cook-function(expression) -> bytes
```

6.5.2 cook-indices

Cook indices into Idenso's canonical structure.

Parameters

```
cook-indices(expression) -> bytes
```

6.5.3 expand-bis

Selectively expand bispinor structures.

Parameters

```
expand-bis(expression) -> bytes
```

6.5.4 expand-metrics

Expand all supported metric structures.

Parameters

```
expand-metrics(expression) -> bytes
```

6.5.5 expand-mink

Selectively expand Minkowski structures.

Parameters

```
expand-mink(expression) -> bytes
```

6.5.6 expand-mink-bis

Selectively expand combined Minkowski and bispinor structures.

Parameters

```
expand-mink-bis(expression) -> bytes
```

6.5.7 list-dangling

Return the dangling indices as compatible Atom payloads.

Parameters

```
list-dangling(expression) -> array
```

6.5.8 simplify-metrics

Contract and simplify metric tensors.

Parameters

```
simplify-metrics(expression) -> bytes
```

6.5.9 to-dots

Rewrite supported contractions as dot products.

Parameters

```
to-dots(expression) -> bytes
```

6.5.10 wrap-dummies

Wrap dummy indices under the given header symbol.

Parameters

```
wrap-dummies(
  expression,
  header
) -> bytes
```

6.5.11 wrap-indices

Wrap all indices under the given header symbol.

Parameters

```
wrap-indices(
  expression,
  header
) -> bytes
```

6.6 Dirac and color transformations

6.6.1 dirac-adjoint

Take the Dirac adjoint of a tensor expression.

Parameters

```
dirac-adjoint(expression) -> bytes
```

6.6.2 expand-color

Selectively expand color structures.

Parameters

```
expand-color(expression) -> bytes
```

6.6.3 simplify-color

Simplify color tensors.

Parameters

```
simplify-color(expression) -> bytes
```

6.6.4 simplify-gamma

Simplify gamma-matrix expressions.

Parameters

```
simplify-gamma(expression) -> bytes
```

6.7 Advanced configuration

6.7.1 init

Create an independent Tydenso API.

The returned dictionary contains tensor constructors, Spenso-aware printers, CBOR inspection, and Idenso transformations. Use a custom `source` only when developing another compatible plugin; ordinary documents can use the imported top-level functions.

```
#let tensors = init()
#let V = (tensors.mink)(4)
#let p = (tensors.vector)("p")
#(tensors.to-typst)(p((tensors.slot)(V, "mu")))
```

p^{μ}

Parameters

```
init(
  namespace: str,
  source: str bytes none,
  grammar: dictionary
) -> dictionary
```

6.7.1.0.1 namespace `str`

Default namespace for symbols parsed from Typst math.

Default: `"spenso"`

6.7.1.0.2 source `str` or `bytes` or `none`

WebAssembly plugin path or uncompressed module bytes. `none` selects the bundled compressed Tydenso engine.

Default: `none`

6.7.1.0.3 grammar `dictionary`

Parser grammar used by `math` unless it receives an explicit override.

Default: `_default-grammar`

7 Compatibility and licensing

This manual describes Tydenso 0.1.0 and Typst 0.14 or newer. Tydenso's original source is released under the MIT License. The bundled engine also contains Spenso, Idenso, and Symbolica; their own upstream terms continue to apply. In particular, the MIT license for this interface does not relicense Symbolica. Read Symbolica's license before redistributing or deploying the WebAssembly bundle.

8 Acknowledgements

Tydenso would not exist without Symbolica, the exact algebra engine beneath its Atom representation. Thank you to Symbolica's contributors, and to the GammaLoop contributors for Spenso and Idenso, whose tensor model, printers, and identities define this package's mathematics.

Thanks also to Tidy, which powers the worked examples and generated reference, and to Parsely, which parses Typst math while preserving the semantic Atom metadata.